



An introduction to the Item Response Warehouse (IRW): A resource for enhancing data usage in psychometrics

Benjamin W. Domingue¹ · Mika Braginsky¹ · Lucy Caffrey-Maffei¹ · Joshua B. Gilbert² · Klint Kanopka³ · Radhika Kapoor¹ · Hansol Lee¹ · Yiqing Liu¹ · Savira Nadela¹ · Guanzhong Pan⁴ · Lijin Zhang¹ · Susu Zhang⁵ · Michael C. Frank¹

Accepted: 2 August 2025 / Published online: 5 September 2025
© The Author(s) 2025

Abstract

The Item Response Warehouse (IRW) is a collection and standardization of a large volume of item response datasets in a free and open-source platform for researchers. We describe key elements of the data standardization process and provide a brief description of the over 900 datasets in the current iteration of the IRW (version 28.2). We describe how to access the data through both the website and an API, and offer a brief tutorial with example R code illustrating how to download data from the IRW and use it in standard psychometric analyses. While we are continuing to develop the IRW, this presentation may help researchers utilize data from this resource for work in psychometrics and related fields.

Keywords Psychometrics · Big data · Item response theory · Measurement · Factor analysis

Introduction

The Item Response Warehouse (IRW) is a large collection of publicly available psychometric datasets standardized to a common format for the purposes of providing data for future research on psychometric methods. This effort is aligned with the growth of standardized data resources in other disciplines that have supercharged subsequent research. For example, in computer science, the introduction of ImageNet (Deng et al., 2009), a large-scale dataset of images, revolutionized AI by providing a critical benchmark for training and evaluation.

While the IRW uses previously available data, the articulation of a data standard and harmonization of data resources to meet that standard is a useful development for both methodologists and applied researchers in psychometrics, mathematical psychology, and related disciplines in the

social and behavioral sciences. Similar data standards have been adopted in other fields such as genetics and neuroscience (Purcell et al., 2007; Gorgolewski et al., 2016), and we propose that the IRW would be equally useful in psychometric research. Further, making these data resources available in a single location with accompanying software increases the degree to which they are findable, accessible, interoperable, and reusable (i.e., FAIR; Wilkinson et al., 2016).

The data resources of the IRW are available via both a website and programmatic access via API. We describe the types of data we include and the structure we use for standardization. The IRW already contains over 2 billion item responses in a standardized format. While some of this data is idiosyncratic, large tranches of it could be easily analyzed in widely used psychometric software such as *mirt*, *lme4*, *lavaan*, *brms*, *psychometrics*, and others (Chalmers, 2012; Rosseel, 2012; Bates et al., 2014; Bürkner, 2021; Epskamp and Rcpp, 2020; Isvoranu et al., 2022); we illustrate how this can be done with brief tutorials. We also describe a number of metadata elements that we have developed, allowing users to filter through the various IRW datasets to identify those with characteristics relevant to them. While we aim to continue to grow the IRW and to increase interoperability with other computational tools, it is already at a stage where it could help improve the rigor and impact of current research.

✉ Benjamin W. Domingue
ben.domingue@gmail.com

¹ Stanford University, Stanford, CA, USA

² Harvard University, Cambridge, Massachusetts, USA

³ New York University, New York, USA

⁴ The University of Edinburgh, Edinburgh, UK

⁵ University of Illinois Urbana-Champaign,
Urbana-Champaign, Illinois, USA

The IRW

The IRW consists of both a collection of ‘tables’ and a system of technological infrastructure that facilitates their acquisition.¹ We first describe the data resources before a description of how users can access the IRW. Our discussion of the IRW begins with a consideration of what data we focus on for inclusion in the IRW.

Inclusion criteria

There is a wide range of data available in the IRW derived from a wide variety of sources. For example, we have included numerous data sets from popular R packages focused on psychometrics (e.g., Mair, 2018; Robitzsch and Steinfeld, 2018; Van der Ark, 2012; George et al., 2016; Robitzsch et al., 2022b; Heller and Wickelmaier, 2013; Robitzsch, 2022a). We also include data from papers that have made data publicly available, educational technology firms that have posted data, etc. Our aim is to collect as much relevant data as possible, and to that end, we have a list of data resources that we plan to target in future collection efforts.

One critical issue that we have aimed to manage in building the IRW pertains to data licensing. In particular, the IRW complies with the licensing requirements under which the data were originally made available. Alongside information about licenses, we also describe the provenance of data (along with tools on the website meant to ease the process of citing the original data) and provide documentation linking the IRW-compliant version of the dataset that we have produced. Finally, the code used to produce IRW-compliant versions of the dataset is publicly available.²

Inclusion criteria for IRW are quite broad and nonspecific. This is by design, given that there are common approaches available for analyzing and understanding a broad range of psychometric data types (Takane and De Leeuw, 1987; Zhang and Chen, 2022). Making more expansive sets of data available to researchers will be beneficial so long as the IRW is structured so that researchers can easily select subsets of data that are relevant for their purposes. For inclusion in the IRW, the core requirement is that the data contain cross-classified responses. By responses, we emphasize that the IRW contains the constituent elements of a psychological measure – the item responses – rather than just the aggregate. For example, data containing only the scale scores of students on some standardized test would not be admissible; the data would need to contain the item responses from which those scale scores were computed. Further, note that the responses

are required to be (1) real (the IRW does not contain simulated data)³ and (2) pertaining to a latent variable or other psychological construct. We thus exclude datasets that solely consist of measures of manifest variables, such as physical attributes (e.g., length) or biomarkers (e.g., heart rate).

The second key requirement is that responses be cross-classified. Responses need to be classified by both a (1) measurement object (e.g., individuals) and a (2) measurement instrument (e.g., items; note that we require that there be more than one item used, so single-item scales are not included). This basic template captures a broad variety of data related to educational testing, survey data including personality and health surveys and Likert-style responses (Likert, 1932), and many other settings. Data that meet these requirements are the key inputs for a wide range of latent variable models in psychometrics and related fields, including item response theory (van der Linden, 2016), Bayesian knowledge tracing (Pelánek, 2017), generalizability theory (Shavelson et al., 1989), factor analysis (Gorsuch, 2014), structural equation modeling (Bowen and Guo, 2011; Kline, 2023), network psychometrics (Epskamp et al., 2017; Isvoranu et al., 2022), and multilevel or cross-classified modeling (Rabe-Hesketh and Skrondal, 2008), among others.

While the canonical case in psychometrics is that of persons responding to items, there are other scenarios that we include in the IRW. For example, legislators’ voting records on individual bills are of interest for creating indices of political partisanship and would meet our data requirements. Similarly, multiple judges scoring the performance of an athlete in a figure skating or gymnastics competition are eligible (given that responses for a single athlete are cross-classified across raters). In many instances, a person is the focus of measurement, but this need not always be the case. For example, persons (acting as the instruments) rating faces (the object of measurement) are included (Chen et al., 2021). We emphasize the importance of persistent identifiers. For example, movie reviews (in the form of a number of stars) from multiple individuals (where the movie is the object being measured and the persons are the instrument) would not meet our criteria unless the reviews contain a persistent identifier of the rater.

The IRW’s breadth in including such a variety of data is by design. This breadth means that the IRW needs to allow users to carefully select tables meeting certain specifications (e.g., users may wish to exclude data that contains repeated trials of an item). However, we argue that this richness of data will ultimately prove beneficial. Larger volumes of data mean that an end user has more options if they want to study some technique in a wide variety of datasets. Further, there are good

¹ A note about nomenclature: we use ‘dataset’ to refer generically to data and ‘table’ to refer to the specific reformatted data resources available via the IRW.

² <https://github.com/itemresponsewarehouse/misc/tree/main/data>

³ For clarity, note that we have begun to construct a separate repository of synthetic/simulated item response data resources, but they are not part of the original IRW and we do not discuss them here.

reasons to suspect that the introduction of non-standard data may lead to methodological innovations via ‘borrowing’. For example, as educational measurement becomes more focused on the problems associated with the measurement of ability in longitudinal systems, we may glean insights using techniques from related fields. As an illustration of how such borrowing may work, techniques used to understand the evolving ideologies of judges (Martin and Quinn, 2002) may offer a methodological template for analysis outside of political science.

The IRW data standard

Once data have been deemed eligible for inclusion in the IRW, they are processed to ensure compliance with the IRW data standard. This data standard is described in detail at the IRW website; we emphasize the key elements here.⁴ After describing these key elements, we turn to a variety of ancillary data that we have also included in a standardized format when available.

We note one important fact about our processing of IRW data. We have generally worked to create tables for the IRW that focus on single constructs. For example, if a single study separately measures anxiety and depression, this will result in two tables in the IRW. Some researchers may wish to further split the tables in the IRW, and some may wish to combine tables.⁵

Additional data elements

While the IRW contains a wide variety of data, all tables contain a few core data elements as a function of our inclusion criteria. The core data elements of the IRW are threefold; we offer a small illustration in Table 1. First, an identifier focusing on the unit that the instrument is attempting to measure (the object of measurement, frequently a person). Second, an identifier of the measurement instrument (frequently an item). Third, a response that is scored so that it can be treated as ordinal. In many cases, this is a matter of coding correct or affirmative responses as 1 and incorrect or disaffirming responses as 0. Many other datasets involve scores, such as the number of points a student might get for a partial credit item. In some cases, the response may be continuous (e.g., the user-selected position of a slider bar). No matter the nature of the response, we have taken care to ensure that it is coded so that the values are meaningful (e.g., numeric values can be treated as ordinal). Note also that, as collected in IRW, the data are in ‘long’ form and each row contains informa-

Table 1 Basic long structure of IRW data

id	item	resp
1	1	0
2	1	1
1	2	0
2	2	1

tion about a single item response.⁶ The elements of Table 1 are sufficient for many forms of analysis, but in other cases, additional data will be available. We turn now to a discussion of these ancillary data elements.

Other key ingredients

Alongside the elements of Table 1, we have included other select data when available. In general, these data elements are specific to the dataset and additional information can be obtained from the references in the IRW dictionary. However, some data elements are sufficiently common to merit standardized formatting. We comment on those here (with their name in the standardized tables in parentheses):

- Response time (*rt*): The reported time to produce an item response, when available, is coded in seconds.
- Date (*date*): The calendar time at which a response was produced is included. Coding of this variable is done in two ways. In some cases, there were only relative dates (e.g., 30 days into data collection). In that case, we convert to the number of seconds since the first piece of data in the dataset was collected). In cases where more exact information was given (e.g., 1:30 PM 03/04/2008), we convert to Unix time (i.e., seconds since JAN 01 1970. (UTC)).⁷ In both cases, the date is available for analysis as a straightforward numeric quantity; in particular, differences and order would be easily ascertainable.
- Rater (*rater*): In scenarios wherein the focal unit is being rated by other observers, an identifier associated with the rater that produced the response is recorded so that rater effects can be studied.
- Q Matrix loadings (*qmatrix_**): When items are classified into a small number of skills (i.e., for the purposes of cognitive diagnostic modeling; Ravand and Robitzsch, 2015), we have included these item-level classifications.
- Treatment indicators (*treat*): When data come from a study with some causal focus, we indicate responses from treated subjects with values of 1 (and control as 0).

⁴ <https://itemresponsewarehouse.org/standard.html>

⁵ For this latter case, we have developed resources to help them do so, see `irw::irw_merge()` in the R package we describe in Section “Programmatic work with IRW data in R”.

⁶ For comparison, a ‘wide’ representation of this simple toy data might be $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ where the columns are items and the rows represent the subject of measurement.

⁷ <https://www.unixtimestamp.com/>

- **Wave (wave):** Wave is recorded in the cases of longitudinal data collection that is punctuated and not continuous (in which case, `date` would be more appropriate).
- **Covariates (cov_*):** Covariates associated with `id`, such as age and gender, are flagged so that they are available for study of, for example, differential item functioning (Camilli, 2006).

In some cases, the IRW contains relatively few tables that contain these additional elements, but we aim to target these areas for future growth. Note that the data standard is evolving as we are identifying new features of data that can be standardized in the IRW.

IRW metadata

We track a variety of metadata elements to help researchers identify and select tables of interest. We describe these separately as quantitative (which are metadata elements that are calculated based on the tables themselves) and qualitative (which are based on human annotation) metadata elements. The quantitative metadata are straightforward; for example, the number of responses, the number of measurement probes, and the number of objects that are measured. We also compute density (as defined below) for each table, as different values of density often require specialized handling. The qualitative metadata describe the measurement tool (e.g., is the tool measuring a cognitive construct? an affective construct?) and the sample (e.g., when the objects being measured are humans, what is the age range of these respondents?). The metadata are critical in that they allow IRW users to rapidly filter the large set of IRW tables down to the specific ones that meet their requirements (e.g., using `irw::irw_filter()`, see Section “[Programmatic work with IRW data in R](#)”).

Using the IRW

The IRW data resources are available via several mechanisms. We first describe the IRW website, which is a key component that documents the data standard and resources, contains information on citations and licenses, and offers some limited tools for querying the data. We then describe an R package designed to expedite work with IRW data.

The IRW website

The IRW website⁸ is built on a stack designed to facilitate data-sharing (Braginsky et al., [In prep](#)) and used previously for other data-sharing efforts (Frank et al., 2017; Sanchez

et al., 2019; Zettersten et al., 2023). We describe the key features of the stack below.

- The IRW website is written in Quarto (<https://quarto.org/>). This platform allows for built-in Jupyter notebooks, which allows for us to build capacity for analysis directly into the website.
- We use Observable (<https://observablehq.com/>) for data visualizations. Observable allows us to produce custom visualizations with interactive elements to enable users to explore elements of the IRW directly on the website.
- The website is hosted via GitHub Pages (<https://pages.github.com/>). This platform allows for automatic rebuilding of the website upon changes to the underlying codebase.
- The data backend for the IRW website is Redivis (<https://redivis.com/>). Given that Redivis offers a suite of API tools, users can both download individual tables as needed and also take advantage of programmatic access to the full data resources of the IRW in common statistical computing environments such as R or Python. We discuss the API in more detail below. Redivis also manages data versioning, an essential feature for purposes of replicability.

This stack allows us to share the data in a way that is meant to be user-friendly. For example, users can directly browse through key summaries of individual tables (as well as the response-level data) using the website and then download those of interest.

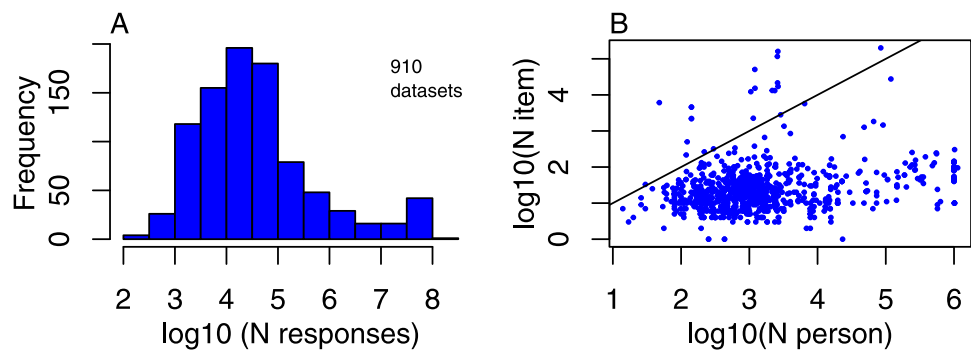
Programmatic work with IRW data in R

To support programmatic access to IRW tables, we provide an R package, `irw`. Key functions include:

- `irw_filter()`: This function can be used to identify tables with specific features (e.g., response time) or based on specific metadata values.
- `irw_fetch()`: Once a set of tables of interest has been identified, they can be retrieved by name using this function; tables will be returned as tibbles (Wickham et al., 2019).
- `irw_long2resp()`: Many use cases for IRW tables will require that the data be reformatted as a ‘wide’ matrix (with rows representing the object being measured and the columns the measurement probes). To facilitate this translation, we have created this function, which is meant to also provide some security so that, for example, responses that come from repeated waves of data collection are handled consistently.
- `irw_save_bibtex()`: We strongly encourage that researchers cite the tables that they use. To facilitate this, this function produces citations based on table names.

⁸ <https://itemresponsewarehouse.org/>

Fig. 1 Overview of the size of the IRW. **A** A histogram showing the frequency of the $\log_{10}$ of the number of item responses in each table. **B** A scatter plot (in $\log_{10}$ scale) showing the number of persons and items in each table. The black line is 45 degree line



An example workflow illustrating these functions is shown below. Full documentation, including reference pages for all functions and usage examples, is available.⁹

A key point of variation among the IRW tables concerns density. Suppose Table 1 is translated into a conventional or ‘wide’ item response table where rows represent people and

Example usage of irw for filtering, fetching, and citing IRW tables

```
# Installing and loading the irw package
devtools::install_github("itemresponsewarehouse/Rpkg")
library(irw)
# Filtering tables with response time (rt) and at least 1000 responses
irw_filter(var = "rt", n_responses = c(1000, Inf))
# Fetching a table and converting to response matrix
df <- irw_fetch("gilbert_meta_2")
resp <- irw_long2resp(df)
# Saving BibTex entries for the given table
irw_save_bibtex("gilbert_meta_2", output_file = "refs.bib")
```

A snapshot of the current data resources of the IRW

As of Version 28.2, the IRW contains 910 tables. A descriptive overview of the sizes of data represented in this version of the IRW is shown in Fig. 1 Panel A; the facts about the IRW that we describe here will evolve as more data is added to the IRW. This panel shows a histogram of the number of responses across all current IRW tables. Some tables are quite small, containing around $10^2 = 100$ responses. Most are much bigger: the median table contains approximately $10^{4.4} \approx 24,000$ responses. Panel B shows a scatter plot of the number of persons (i.e., `id` identifiers) by the number of items (i.e., `item` identifiers). The majority of the tables involve fewer items than measurement objects, but there are a few tables containing very large numbers of items. Note that we typically limit tables to one million respondents (based on a random sample from the full set); if researchers are interested in the complete datasets, they can use the provided code that reformats the data to construct an IRW-compatible table with all respondents (which could then be analyzed alongside other IRW data resources).

columns represent items. In the simplest case, every person responds to every item; this scenario is relatively common in our data. We index density by the proportions of cells in such a table that are not empty; in this simple case, density is 1 (as 100% of the cells have valid responses). In many cases, individuals do not respond to all items, and density may be less than 1. Common reasons for such density might be due to respondent behavior – e.g., respondents skipping items – or due to assessment design – e.g., balanced incomplete booklets (wherein respondents are only meant to take a small number of items) or adaptive assessments (in which case external information on items is typically required to obtain accurate ability estimates). Of the tables in the IRW, 27% have density values less than 1, but only 10% of these tables have density values less than .5. In other cases, respondents may attempt items multiple times, leading to density greater than 1. This commonly occurs in psychological testing when respondents make repeated attempts at a trial (especially when response time is the critical outcome rather than accuracy alone) and intensive longitudinal or experience sampling designs (Hamaker and Wichers, 2017; Asparouhov and Muthén, 2020; McNeish et al., 2021; Stieger and Kuhlmann, 2018). Of the current IRW tables, 24% have density values greater than 1.

⁹ <https://itemresponsewarehouse.github.io/Rpkg/>

Brief tutorial for accessing and analyzing the IRW in R

The web interface can be used to get some understanding of the data in the IRW and to download individual files of interest. Larger-scale work, however, may benefit from alternative means of accessing the IRW data. In particular, heavy users of the IRW will benefit from the use of the custom R package `irw` for working with IRW data. In particular, we are incorporating a variety of techniques for filtering and fetching data from the IRW in this package.

Standard psychometric analysis

Below, we demonstrate how straightforward analysis can be performed on a sample table (taken from Kim et al., 2023). We conduct analysis using functions from a variety of commonly used R packages (Chalmers, 2012; Dinno, 2018; Revelle, 2024) to compute reliability, conduct dimensionality analysis, and then estimate factor analysis and IRT models.

Standard psychometric analysis of sample table

```
df<-irw::irw_fetch("gilbert_meta_2") ##fetch data from the IRW
df$item <- paste("item_", df$item, sep = "")
resp <- irwpkg::irw_long2resp(df) ##reformat long data to wide
resp$id <- NULL
psych::alpha(resp) ##compute cronbach's alpha
paran::paran(resp) ##parallel analysis to estimate dimensionality
psych::fa(cov(resp), nfactors = 3) ##exploratory factor analysis
mirt::mirt(resp, 1, "Rasch") ##fitting the Rasch model
```

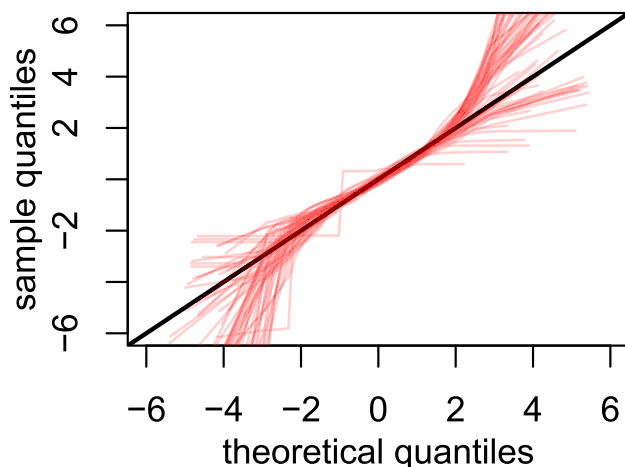


Fig. 2 Lognormality of response time data from $N = 68$ tables in the IRW that contain response time

Response time analysis

We now turn to a more complex example.¹⁰ In this example, we first identify all data with response time (i.e., a `rt` column), download the data, and analyze response times. This illustration could serve as a template for how to identify specific types of data in the IRW. We analyze 68 tables with response time in the IRW.¹¹

We conduct a straightforward descriptive analysis of this data: we consider the degree to which the response time data for all persons and items appear lognormal. The lognormal model (e.g., a response of 10s is transformed to $\log(10) = 2.3$) is a popular model for response time (van der Linden, 2006). Rather than formal tests (Sinharay and van Rijn, 2020) assessing whether the response times are lognormal, we focus on descriptive analyses (after restricting analysis to responses less than 30 minutes) using QQ plots, see Fig. 2. In general, the lognormal assumption seems to match the data fairly well through roughly the $z \in \{-2, 2\}$ range. Outside of that range, there is clearly variation in the kurtosis of many distributions that would be inconsistent with the lognormal assumption.

Analysis of item-level treatment effects

There has been recent interest in analyzing the degree to which there is variation in item-level treatment effects of

¹⁰ Code available at https://github.com/itemresponsewarehouse/misc/blob/main/manuscript_src/02_responsetime.R.

¹¹ Data come from: Jones et al. (2024); Artistic (2023); Jorgenson (2016); Yarkoni (2010); Van Der Maas and Wagenmakers (2005); Cizek and Wollack (2016); Borst et al. (2011); Depression (2021); Sug-gate (2024); Settles (2018a, b, c, d, e, f, g, h, i); Exposure (n.d.); Belfi and Kacirek (2020); Goldberg (1992a, b, c, d, e); Brown et al. (2013); Hampshire et al. (2012a, b, c); Brotherton et al. (2013); Papoušek et al. (2016); Glatz (2023a, b); Jastrzębski et al. (2022); Zhang et al. (2023); Development (n.d.); Inoue et al. (2024); Christie and Geis (1970); Ratcliff and Hendrickson (2021a, b); Multifactor (n.d.); O'Brien and Yeatman (2020); Ratcliff and Hendrickson (2021c, d, e, f); Much et al. (2025); Nisbet and Zelenski (2013); Hooper et al. (2024); Archambeau et al. (2022); Richmond et al. (2003); Bohn et al. (2023); Pisa (2017a, b, c, 2019a, 2018cb, 2019c, d, 2023a, b, c); Mirels and Garrett (1971); Yeatman et al. (2021); Ratcliff and Rouder (1998); Bendall et al. (2024); Duñabeitia (2021); Bohn et al. (2022).

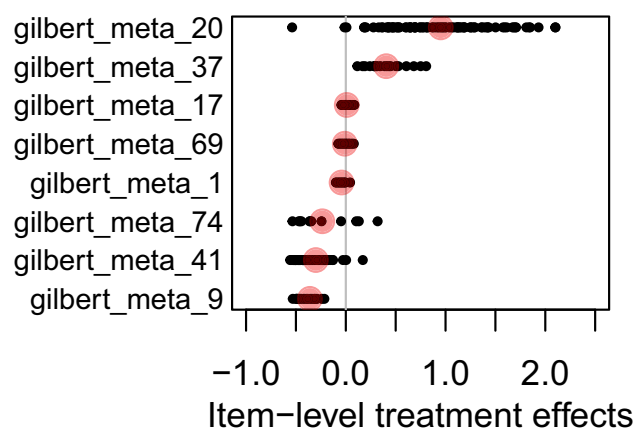


Fig. 3 Item-level heterogeneous treatment effects on the logit scale: Overall treatment effect in red with item-level effects shown as black dots

randomized experiments (Ahmed et al., 2023; Gilbert et al., 2023b, 2024c,d; Halpin and Gilbert, 2024; Gilbert et al., 2024e; Sales et al., 2021). Data in the IRW from such experiments have a `treat` column that indicates whether a respondent with a given `id` was a member of a treatment group. We offer a brief example of such analysis using 8 RCT tables (Banerjee et al., 2017; Carpena, 2024; Cilliers, 2023; Duflo et al., 2024; Gersten et al., 2019; Gilbert, 2023a; Gilbert et al., 2024a; Hidrobo et al., 2016).¹² As in earlier work (Ahmed et al., 2023; Gilbert et al., 2024c), we demonstrate pronounced variation across RCT outcomes in item-level heterogeneity with respect to treatment sensitivity (Fig. 3). In one table (`gilbert_meta_69`, Banerjee et al., 2017), we see quite limited variation. In this case, the effect caused by the treatment is fairly consistent across the outcome's items. In other cases (e.g., `gilbert_meta_20`, Gersten et al., 2019), there is clear variation around a general effect of treatment. At present, the IRW currently contains over 95 tables from RCTs, which may be useful for research synthesizing causal inference and psychometric methods.

Discussion

With the IRW, we aim to foster the development of a community resource that contains standardized and accessible item response datasets. In this paper, we have described the data available in the IRW, how it is standardized, and given examples of how to use and analyze this data. While development is ongoing, there is already a large volume of tables available to researchers in the IRW.¹³ These data are standardized,

have information related to reuse and citation readily available, and can be pulled into the computing environments used by many psychometrics researchers.

There are limitations of the IRW data. The datasets used here should be viewed as potentially compromised due to choices made for the purposes of standardization. Decisions made for standardization and coherent analysis, given the generic interests of IRW researchers, may depart from the original intent of the data source (e.g., in general, we made decisions about how to define 'items' when there was some ambiguity that resulted in relatively few items rather than relatively more). While data in the IRW are easily accessible, they may not be authoritative, and researchers should consult the primary sources for each data source to verify the interpretation of the data (e.g., the work in Gilbert et al. (2025) provides one example of how this could be done, as the authors supplement data drawn from the IRW with replication materials from the original sources to explore item wording effects). We view the accessible standardized versions of these datasets in the IRW as an important resource despite this limitation.

Our ultimate goal is to make the IRW easy to use for researchers in psychometrics, psychology, educational measurement, and behavioral research. We aim to make it convenient to readily consider the performance of a hypothetical psychometric model in a wide array of data. Being able to readily probe variation in performance across a variety of datasets will make the generalizability of a given model's use cases much more immediately apparent. Being able to consider such questions will, we hope, both push research in more productive directions and also lead to novel research questions.

Author Contributions This project was conceptualized by BWD with assistance from KK and MCF. The webpage was designed by MB and MF with contributions from BWD and GP. Design of the data standard was led by BWD with input from LCM, JBG, KK, RK, GP, SZ, and MF. Key data contributors include BWD, LCM, JBG, and GP, with additional contributions from KK, RK, SN, and YL. Work on the `irw` was led by HL. Analysis was done by BWD. The manuscript was written by BD with input from all authors.

Funding Funding for the IRW was provided by the Stanford Office of the Vice Provost and Dean of Research (VPDoR), the Jacobs Foundation, and the Institute of Education Sciences, U.S. Department of Education, through Grant R305D240025 to Stanford University. The opinions expressed are those of the authors and do not represent the views of the Institute or the U.S. Department of Education.

Data Availability See <https://itemresponsewarehouse.org>.

Code Availability Relevant materials are available at <https://github.com/itemresponsewarehouse>.

Declarations

Conflicts of Interest The authors report no conflicts of interest.

¹² Code available at https://github.com/itemresponsewarehouse/misc/blob/main/manuscript_src/03_il-hte.R.

¹³ Note that we welcome contributions to the IRW and describe how users can contribute data to the IRW at: <https://itemresponsewarehouse.org/contribute.html>.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Open practices statement The data and materials for all experiments are available.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Ahmed, I., Bertling, M., Zhang, L., Ho, A. D., Loyalka, P., Xue, H., Rozelle, S., & Domingue, B. W. (2024). Heterogeneity of item-treatment interactions masks complexity and generalizability in randomized controlled trials. *Journal of Research on Educational Effectiveness*, 1-22.
- Archambeau, K., Couto, J., & Van Maanen, L. (2022). Non-parametric mixture modeling of cognitive psychological data: A new method to disentangle hidden strategies. *Behavior Research Methods*, 55 (5), 2232-2248. Retrieved from <https://doi.org/10.3758/s13428-022-01837-0>.
- Artistic preferences scale [dataset and codebook] [Dataset]. (2023). Retrieved from http://openpsychometrics.org/_rawdata/APS_data.zip.
- Asparouhov, T., & Muthén, B. (2020). Comparison of models for the analysis of intensive longitudinal data. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(2), 275-297.
- Banerjee, A., Banerji, R., Duflo, E., Kannan, H., Mukerji, S., Shotland, M., & Walton, M. (2017). Raw and replication data for: 'From proof of concept to scalable policies' and 'mainstreaming an effective intervention'. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/DUBA3J>.
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2014). Fitting linear mixed-effects models using lme4. arXiv preprint [arXiv:1406.5823](https://arxiv.org/abs/1406.5823).
- Belfi, A. M., & Kacirek, K. (2020). The famous melodies stimulus set. *Behavior Research Methods*, 53 (1), 34-48. Retrieved from <https://doi.org/10.3758/s13428-020-01411-6>.
- Bendall, R. C. A., Royle, S., Dodds, J., Watmough, H., Gillman, J. C., Beevers, D., & Gregory, S. E. A. (2024). The salford nature environments database (sned): An open-access database of standardized high-quality pictures from natural environments. *Behavior Research Methods*, 57 (1). Retrieved from <https://doi.org/10.3758/s13428-024-02556-4>.
- Bohn, M., Prein, J., Koch, T., Bee, R. M., Delikaya, B., Haun, D., & Gagarina, N. (2023). Orev: An item response theory-based open receptive vocabulary task for 3- to 8-year-old children. *Behavior Research Methods*, 56 (3), 2595-2605. Retrieved from <https://doi.org/10.3758/s13428-023-02169-3>.
- Bohn, M., Prein, J. C., Koch, T., Bee, R. M., Delikaya, B., Haun, D., & Gagarina, N. (2022). Orev: An item response theory based open receptive vocabulary task for 3 to 8-year-old children. Retrieved from <https://doi.org/10.31234/osf.io/4z86w>.
- Borst, G., Kievit, R. A., Thompson, W. L., & Kosslyn, S. M. (2011). Mental rotation is not easily cognitively penetrable. *Journal of Cognitive Psychology*, 23 (1), 60-75. Retrieved from <https://doi.org/10.1080/20445911.2011.454498>.
- Bowen, N. K., & Guo, S. (2011). *Structural equation modeling*. Oxford University Press.
- Braginsky, M., Chandler, Z., Mathew, I., Domingue, B. W., & Frank, M. C. (In prep). Datapages to promote interoperable and reusable data sharing (Tech. Rep.).
- Brotherton, R., French, C. C., & Pickering, A. D. (2013). Measuring belief in conspiracy theories: The generic conspiracist beliefs scale. *Frontiers in Psychology*, 4. Retrieved from <https://doi.org/10.3389/fpsyg.2013.00279>.
- Brown, L. L., Acevedo, B., & Fisher, H. E. (2013). Neural correlates of four broad temperament dimensions: Testing predictions for a novel construct of personality. *PLoS ONE*, 8 (11), e78734. Retrieved from <https://doi.org/10.1371/journal.pone.0078734>.
- Bürkner, P.-C. (2021). Bayesian item response modeling in r with brms and stan. *Journal of Statistical Software*, 100, 1-54.
- Camilli, G. (2006). *Test fairness. Educational measurement*, 4, 221-256.
- Carpena, F. (2024). Replication data for: Entertainment-education for better health: Insights from a field experiment in india. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/EJ03JR>.
- Chalmers, R. P. (2012). mirt: A multidimensional item response theory package for the r environment. *Journal of statistical Software*, 48, 1-29.
- Chen, J. M., Norman, J. B., & Nam, Y. (2021). Broadening the stimulus set: Introducing the american multiracial faces database. *Behavior Research Methods*, 53, 371-389.
- Christie, R., & Geis, F. L. (1970). *Studies in machiavellianism*. Academic Press. Retrieved from https://openpsychometrics.org/_rawdata/.
- Cilliers, J. (2023). Replication data for: Reading skills transfer best from home language to a second language: Policy lessons from two field experiments in south africa. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/RSUADW>.
- Cizek, G. J., & Wollack, J. A. (Eds.). (2016). *Handbook of quantitative methods for detecting cheating on tests*. Abingdon, England: Routledge. Retrieved from <https://cran.r-project.org/web/packages/LNIRT/index.html>.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In 2009 IEEE conference on computer vision and pattern recognition (pp. 248-255).
- Depression anxiety stress scales [Dataset]. (2021). Retrieved from http://openpsychometrics.org/_rawdata/DASS_data_21.02.19.zip.
- Development of the multidimensional introversion-extraversion scales [Dataset]. (n.d.). Retrieved from http://openpsychometrics.org/_rawdata/MIES_Dev_Data.zip.
- Dinno, A. (2018). paran: Horn's test of principal components/factors [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=paran> (R package version 1.5.2).
- Duflo, A., Kiessel, J., & Lucas, A. M. (2024, February). Experimental evidence on four policies to increase learning at scale. *The Economic Journal*, 134 (661), 1985-2008. Retrieved from <https://doi.org/10.1093/ej/ueae003>.
- Duñabeitia, J. A. (2021). Supplementary files for the predictors of general knowledge: Data from a spanish megas-

- tudy. figshare. Retrieved from https://figshare.com/articles/dataset/Supplementary_files_for_The_predictors_of_general_knowledge_data_from_a_Spanish_megastudy/14073899.
- Epskamp, S., Epskamp, M. S., & Rcpp, L. (2020). Package 'psychonetrics'. Computer software.
- Epskamp, S., Rhemtulla, M., & Borsboom, D. (2017). Generalized network psychometrics: Combining network and latent variable models. *Psychometrika*, 82, 904–927.
- Exposure based face memory test [dataset and codebook]. (n.d.). <http://openpsychometrics.org/rawdata/EBFMT.zip>.
- Frank, M. C., Braginsky, M., Yurovsky, D., & Marchman, V. A. (2017). Wordbank: An open repository for developmental vocabulary data. *Journal of child language*, 44(3), 677–694.
- George, A. C., Robitzsch, A., Kiefer, T., Groß, J., & Ünlü, A. (2016). The R package CDM for cognitive diagnosis models. *Journal of Statistical Software*, 74(2), 1–24.
- Gersten, R., Jayanthi, M., Dimino, J., & Karp, K. (2019). Impact of a tier 2 fractions intervention on 5th grade at-risk students' fractions achievement. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/Z4WMFE>.
- Gilbert, J. (2023a). Replication data for: Modeling item-level heterogeneous treatment effects with the explanatory item response model: Leveraging large-scale online assessments to pinpoint the impact of educational interventions. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/QARRYT>.
- Gilbert, J., Kim, J., & Miratrix, L. (2024a). Replication data for: Leveraging item parameter drift to assess transfer effects in vocabulary learning. Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/ZFILKZ>.
- Gilbert, J. B., Hieronymus, F., Eriksson, E., & Domingue, B. W. (2024d). Item-level heterogeneous treatment effects of selective serotonin reuptake inhibitors (ssris) on depression: Implications for inference, generalizability, and identification. *Epidemiologic Methods*, 13(1), 20240006.
- Gilbert, J. B., Himmelsbach, Z., Soland, J., Joshi, M., & Domingue, B. W. (2024c). Estimating heterogeneous treatment effects with item-level outcome data: Insights from item response theory. <https://doi.org/10.1002/pam.70025>.
- Gilbert, J. B., Kim, J. S., & Miratrix, L. W. (2023b). Modeling item-level heterogeneous treatment effects with the explanatory item response model: Leveraging large-scale online assessments to pinpoint the impact of educational interventions. *Journal of Educational and Behavioral Statistics*, 10769986231171710.
- Gilbert, J. B., Kim, J. S., & Miratrix, L. W. (2024). Leveraging item parameter drift to assess transfer effects in vocabulary learning. *Applied Measurement in Education*, 37(3), 240–257.
- Gilbert, J. B., Zhang, L., Ulitzsch, E., & Domingue, B. W. (2025). Polytomous explanatory item response models for item discrimination: Assessing negative-framing effects in social-emotional learning surveys. *Behavior Research Methods*, 57.
- Glatz, T. (2023a). Open data of a cluster-randomized trial of graphogame-nl in groningen, the netherlands and ghent, belgium. Open Science Framework. Retrieved from <https://osf.io/4p8hz/>.
- Glatz, T. (2023b). Open data of a cluster-randomized trial of graphogame-nl in groningen, the netherlands and ghent, belgium. Open Science Framework. Retrieved from <https://osf.io/4p8hz/>.
- Goldberg, L. R. (1992a). The development of markers for the big-five factor structure. *Psychological Assessment*, 4(1), 26–42. Retrieved from <https://doi.org/10.1037/1040-3590.4.1.26>.
- Goldberg, L. R. (1992b). The development of markers for the big-five factor structure. *Psychological Assessment*, 4(1), 26–42. Retrieved from <https://doi.org/10.1037/1040-3590.4.1.26>.
- Goldberg, L. R. (1992c). The development of markers for the big-five factor structure. *Psychological Assessment*, 4(1), 26–42. Retrieved from <https://doi.org/10.1037/1040-3590.4.1.26>.
- Goldberg, L. R. (1992d). The development of markers for the big-five factor structure. *Psychological Assessment*, 4(1), 26–42. Retrieved from <https://doi.org/10.1037/1040-3590.4.1.26>.
- Goldberg, L. R. (1992e). The development of markers for the big-five factor structure. *Psychological Assessment*, 4(1), 26–42. Retrieved from <https://doi.org/10.1037/1040-3590.4.1.26>.
- Gorgolewski, K. J., Auer, T., Calhoun, V. D., Craddock, R. C., Das, S., Duff, E. P., Flandin, G., Ghosh, S. S., Glatard, T., Halchenko, Y. O., & Handwerker, D. A. (2016). The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Scientific Data*, 3(1), 1–9.
- Gorsuch, R. L. (2014). Factor analysis: Classic edition. Routledge.
- Halpin, P. F., & Gilbert, J. B. (2024). Testing whether reported treatment effects are unduly dependent on the specific outcome measure used. arXiv preprint.
- Hamaker, E. L., & Wichers, M. (2017). No time like the present: Discovering the hidden dynamics in intensive longitudinal data. *Current Directions in Psychological Science*, 26(1), 10–15.
- Hampshire, A., Highfield, R. R., Parkin, B. L., & Owen, A. M. (2012a). Fractionating human intelligence. *Neuron*, 76(6), 1225–1237. Retrieved from <https://doi.org/10.1016/j.neuron.201206.022>.
- Hampshire, A., Highfield, R. R., Parkin, B. L., & Owen, A. M. (2012b). Fractionating human intelligence. *Neuron*, 76(6), 1225–1237. Retrieved from <https://doi.org/10.1016/j.neuron.2012.06.022>.
- Hampshire, A., Highfield, R. R., Parkin, B. L., & Owen, A. M. (2012c). Fractionating human intelligence. *Neuron*, 76(6), 1225–1237. Retrieved from <https://doi.org/10.1016/j.neuron.2012.06.022>.
- Heller, J., & Wickelmaier, F. (2013). Minimum discrepancy estimation in probabilistic knowledge structures. *Electronic Notes in Discrete Mathematics*, 42, 49–56.
- Hidrobo, M., Peterman, A., & Heise, L. (2016). Replication data for: The effect of cash, vouchers, and food transfers on intimate partner violence: Evidence from a randomized experiment in northern ecuador. ICPSR - Interuniversity Consortium for Political and Social Research. Retrieved from <https://www.openicpsr.org/openicpsr/project/113634/version/V1/view>.
- Hooper, M. A., Tomarken, A., & Gauthier, I. (2024). Measuring visual ability in linguistically diverse populations. *Behavior Research Methods*, 57(1). Retrieved from <https://doi.org/10.3758/s13428-024-02579-x>.
- Inoue, T., Chen, Y., & Ohyanagi, T. (2024). Assessing handwriting skills in a web browser: Development and validation of an automated online test in japanese kanji. Retrieved from <https://osf.io/gver2>.
- Isvoranu, A.-M., Epskamp, S., Waldorp, L., & Borsboom, D. (2022). Network psychometrics with r: A guide for behavioral and social scientists. Taylor & Francis.
- Jastrzębski, J., Ociepa, M., & Chuderski, A. (2022). Graph mapping: A novel and simple test to validly assess fluid reasoning. *Behavior Research Methods*, 55(1), 448–460. Retrieved from <https://doi.org/10.3758/s13428-022-01846-z>.
- Jones, A., Petrovskaya, E., & Stafford, T. (2024). Exploring the multi-verse of analysis options for the alcohol stroop. *Behavior Research Methods*, 56(4), 3578–3588. Retrieved from <https://doi.org/10.3758/s13428-024-02377-5>.
- Jorgenson, & E. (2016). Development of the open hemispheric brain dominance scale [Dataset]. Retrieved from http://openpsychometrics.org/_rawdata/OHBDS-data.zip.
- Kim, J. S., Burkhauser, M. A., Relyea, J. E., Gilbert, J. B., Scherer, E., Fitzgerald, J., Mosher, D., & McIntyre, J. (2023). A longitudinal randomized trial of a sustained content literacy intervention

- from first to second grade: Transfer effects on students' reading comprehension. *Journal of Educational Psychology*, 115(1), 73.
- Kline, R. B. (2023). Principles and practice of structural equation modeling. Guilford publications
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of psychology*.
- Mair, P. (2018). *Modern psychometrics with r*. Springer.
- Martin, A. D., & Quinn, K. M. (2002). Dynamic ideal point estimation via markov chain monte carlo for the us supreme court, 1953–1999. *Political analysis*, 10(2), 134–153.
- McNeish, D., Mackinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2021). Measurement in intensive longitudinal data. *Structural equation modeling: A multidisciplinary Journal*, 28(5), 807–822.
- Mirels, H. L. & Garrett, J. B. (1971). Protestant work ethic scale [dataset and codebook] [Dataset]. Retrieved from http://openpsychometrics.org/_rawdata/PWE_data.zip.
- Much, S., Mutak, A., Pohl, S., & Ranger, J. (2025). Data from a validation study of two psychometric models on test-taking behavior. *Journal of Open Psychology Data*, 13(1), 1. Retrieved from <https://doi.org/10.5334/jopd.124>.
- Multifactor general knowledge test. (n.d.). <https://openpsychometrics.org/tests/MGKT2/>. (Accessed from https://openpsychometrics.org/_rawdata/).
- Nisbet, E. K., & Zelenski, J. M. (2013). The nr-6: a new brief measure of nature relatedness. *Frontiers in Psychology*, 4. Retrieved from <https://doi.org/10.3389/fpsyg.2013.00813>.
- O'Brien, G., & Yeatman, J. D. (2020). Bridging sensory and language theories of dyslexia: Toward a multifactorial model. *Developmental Science*, 24(3). Retrieved from <https://doi.org/10.1111/desc.13039>.
- Papoušek, J., Pelánek, R., & Stanislav, V. (2016). Adaptive geography practice data set. *Journal of Learning Analytics*, 3(2), 317–321. Retrieved from <https://doi.org/10.18608/jla.2016.32.17>.
- Pelánek, R. (2017). Bayesian knowledge tracing, logistic models, and beyond: an overview of learner modeling techniques. *User Modeling and User-Adapted Interaction*, 27, 313–350.
- Pisa 2015 assessment and analytical framework: Science, reading, mathematic, financial literacy and collaborative problem solving. (2017a). OECD. Retrieved from <https://doi.org/10.1787/9789264281820-en>.
- Pisa 2015 assessment and analytical framework: Science, reading, mathematic, financial literacy and collaborative problem solving. (2017b). OECD. Retrieved from <https://doi.org/10.1787/9789264281820-en>.
- Pisa 2015 assessment and analytical framework: Science, reading, mathematic, financial literacy and collaborative problem solving. (2017c). OECD. Retrieved from <https://doi.org/10.1787/9789264281820-en>.
- Pisa 2018 assessment and analytical framework. (2019a). OECD. Retrieved from <https://doi.org/10.1787/b25efab8-en>.
- Pisa 2018 assessment and analytical framework. (2019b). OECD. Retrieved from <https://doi.org/10.1787/b25efab8-en>.
- Pisa 2018 assessment and analytical framework. (2019c). OECD. Retrieved from <https://doi.org/10.1787/b25efab8-en>.
- Pisa 2018 assessment and analytical framework. (2019d). OECD. Retrieved from <https://doi.org/10.1787/b25efab8-en>.
- Pisa 2022 assessment and analytical framework. (2023a). OECD. Retrieved from <https://doi.org/10.1787/df0bf9c-en>.
- Pisa 2022 assessment and analytical framework. (2023b). OECD. Retrieved from <https://doi.org/10.1787/df0bf9c-en>.
- Pisa 2022 assessment and analytical framework. (2023c). OECD. Retrieved from <https://doi.org/10.1787/df0bf9c-en>.
- Purcell, S., Neale, B., Todd-Brown, K., Thomas, L., Ferreira, M. A., Bender, D., Maller, J., Sklar, P., De Bakker, P. I., Daly, M. J., & Sham, P. C. (2007). Plink: A tool set for whole-genome association and population-based linkage analyses. *The American Journal of Human Genetics*, 81(3), 559–575.
- Rabe-Hesketh, S., & Skrondal, A. (2008). Multilevel and longitudinal modeling using stata. STATA press.
- Ratcliff, R., & Hendrickson, A. T. (2021). Do data from mechanical turk subjects replicate accuracy, response time, and diffusion modeling results? *Behavior Research Methods*, 53(6), 2302–2325. Retrieved from <https://doi.org/10.3758/s13428-021-01573-x>.
- Ratcliff, R., & Hendrickson, A. T. (2021). Do data from mechanical turk subjects replicate accuracy, response time, and diffusion modeling results? *Behavior Research Methods*, 53(6), 2302–2325. Retrieved from <https://doi.org/10.3758/s13428-021-01573-x>.
- Ratcliff, R., & Hendrickson, A. T. (2021). Do data from mechanical turk subjects replicate accuracy, response time, and diffusion modeling results? *Behavior Research Methods*, 53(6), 2302–2325. Retrieved from <https://doi.org/10.3758/s13428-021-01573-x>.
- Ratcliff, R., & Hendrickson, A. T. (2021). Do data from mechanical turk subjects replicate accuracy, response time, and diffusion modeling results? *Behavior Research Methods*, 53(6), 2302–2325. Retrieved from <https://doi.org/10.3758/s13428-021-01573-x>.
- Ratcliff, R., & Hendrickson, A. T. (2021). Do data from mechanical turk subjects replicate accuracy, response time, and diffusion modeling results? *Behavior Research Methods*, 53(6), 2302–2325. Retrieved from <https://doi.org/10.3758/s13428-021-01573-x>.
- Ratcliff, R., & Rouder, J. N. (1998). Modeling response times for two-choice decisions. *Psychological Science*, 9(5), 347–356. Retrieved from <https://doi.org/10.1111/1467-9280.00067>.
- Ravand, H., & Robitzsch, A. (2015). Cognitive diagnostic modeling using r. *Practical Assessment, Research, and Evaluation*, 20(1), 11.
- Richmond, V. P., McCroskey, J. C., & Johnson, A. D. (2003). Development of the nonverbal immediacy scale (nis): Measures of self-and other-perceived nonverbal immediacy. *Communication Quarterly*, 51(4), 504–517. Retrieved from <https://doi.org/10.1080/01463370309370170>.
- Robitzsch, A. (2022a). sirt: Supplementary item response theory models [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=sirt> (R package version 3.12-66).
- Robitzsch, A., Kiefer, T., & Wu, M. (2022b). Tam: Test analysis modules [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=TAM> (R package version 4.1-4).
- Robitzsch, A., & Steinfeld, J. (2018). Item response models for human ratings: Overview, estimation methods, and implementation in R. *Psychological Test and Assessment Modeling*, 60(1), 101–139.
- Rosseel, Y. (2012). lavaan: An r package for structural equation modeling. *Journal of Statistical Software*, 48, 1–36.
- Sales, A., Prihar, E., Heffernan, N., & Pane, J. F. (2021). The effect of an intelligent tutor on performance on specific posttest problems. *International Educational Data Mining Society*.
- Sanchez, A., Meylan, S. C., Braginsky, M., MacDonald, K. E., Yurovsky, D., & Frank, M. C. (2019). childes-db: A flexible and reproducible interface to the child language data exchange system. *Behavior Research Methods*, 51, 1928–1941.
- Settles, Burr. (2018a). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018b). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.

- Settles, Burr. (2018c). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018d). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018e). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018f). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018g). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018h). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Settles, Burr. (2018i). Data for the 2018 duolingo shared task on second language acquisition modeling (slam). Harvard Dataverse. Retrieved from <https://dataverse.harvard.edu/citation?persistentId=doi.org/10.7910/DVN/8SWHNO>.
- Shavelson, R. J., Webb, N. M., & Rowley, G. L. (1989). Generalizability theory. *American Psychologist*, 44(6), 922.
- Sinharay, S., & van Rijn, P. W. (2020). Assessing fit of the lognormal model for response times. *Journal of Educational and Behavioral Statistics*, 45(5), 534–568. Retrieved from <https://doi.org/10.3102/1076998620911935>
- Stieger, S., & Kuhlmann, T. (2018). Validating psychometric questionnaires using experience-sampling data: The case of nightmare distress. *Frontiers in Neuroscience*, 12, 901.
- Suggate, S. P. (2024). Beyond self-report: Measuring visual, auditory, and tactile mental imagery using a mental comparison task. *Behavior Research Methods*, 56(8), 8658–8676. Retrieved from <https://doi.org/10.3758/s13428-024-02496-z>
- Takane, Y., & De Leeuw, J. (1987). On the relationship between item response theory and factor analysis of discretized variables. *Psychometrika*, 52(3), 393–408.
- Van der Ark, L. A. (2012). New developments in mokken scale analysis in R. *Journal of Statistical Software*, 48(5), 1–27. Retrieved from <https://www.jstatsoft.org/article/view/v048i05>
- van der Linden, W. J. (2006). A lognormal model for response times on test items. *Journal of Educational and Behavioral Statistics*, 31(2), 181–204.
- van der Linden, W. J. (2016). Handbook of item response theory: Volume 1: Models. CRC press.
- Van Der Maas, H. L. J., & Wagenmakers, E.-J. (2005). A psychometric analysis of chess expertise. *The American Journal of Psychology*, 118(1), 29–60. Retrieved from <https://doi.org/10.2307/30039042>.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., & Kuhn, M. (2019). Welcome to the tidyverse. *Journal of open source software*, 4(43), 1686.
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J. W., da Silva Santos, L. B., Bourne, P. E., & Bouwman, J. (2016). The fair guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1), 1–9.
- Revelle, W. (2024). Psych: Procedures for psychological, psychometric, and personality research [Computer software manual]. Evanston, Illinois. Retrieved from <https://CRAN.R-project.org/package=psych> (R package version 2.4.1).
- Yarkoni, T. (2010). The abbreviation of personality, or how to measure 200 personality scales with 200 items. *Journal of Research in Personality*, 44(2), 180–198. Retrieved from <https://doi.org/10.1016/j.jrp.2010.01.002>.
- Yeatman, J. D., Tang, K. A., Donnelly, P. M., Yablonski, M., Ramamurthy, M., Karipidis, I. I., Caffarra, S., Takada, M.E., Kanopka, K., Ben-Shachar, M., & Domingue, B. W. (2021). Rapid online assessment of reading ability. *Scientific Reports*, 11(1). Retrieved from <https://doi.org/10.1038/s41598-021-85907-x>.
- Zettersten, M., Yurovsky, D., Xu, T. L., Uner, S., Tsui, A. S. M., Schneider, R. M., Saleh, A. N., Meylan, S. C., Marchman, V. A., Mankewitz, J., & MacDonald, K. (2023). Peekbank: An open, large-scale repository for developmental eye-tracking data of children's word recognition. *Behavior Research Methods*, 55(5), 2485–2500.
- Zhang, S., & Chen, Y. (2022). Computation for latent variable model estimation: A unified stochastic proximal framework. *psychometrika*, 87(4), 1473–1502.
- Zhang, S., Wang, S., Chen, Y., & Kwon, S. (2023). hmcdm: Hidden markov cognitive diagnosis models for learning [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=hmcdm> (R package version 2.1.1).

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.